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* Note: $\mathcal{P} = \{P_\theta | \theta \in \Theta\}$ is a collecⁿ of distⁿ based on parameter θ .

* $X \sim P_\theta$ (θ unknown)

$T(X) \rightarrow$ an estimate of θ

$S(X) \rightarrow$ a suff. stat.

Theorem: $T^*(X) = T^*(s) = E(T|S)$ is at least as good an estimate as $T(X)$ in the sense that

(Rao-Blackwell)

$$R(\theta, T) \geq R(\theta, T^*) \quad \forall \theta \in \Theta$$

+ Note: T & T^* have same bias ($E(E(T|S)) = E(T)$)

Proof: $\text{Var}(T) = \text{Var}[E(T|S)] + E[\text{Var}(T|S)]$ (Prob-I stuff)

$$\Rightarrow \text{Var}(T) \geq \text{Var}(T^*)$$

$$\Rightarrow \text{Bias}^2(T) + \text{Var}(T) \geq \text{Bias}^2(T^*) + \text{Var}(T^*)$$

$$(\text{Bias}(T) = \text{Bias}(T^*))$$

$$\Rightarrow R(\theta, T) \geq R(\theta, T^*) \quad \forall \theta \in \Theta \quad \square$$

+ Note: - This process is called Rao-Blackwellization (cut f is it?)

- Inequality strict $\Leftrightarrow E[V(T|S)] > 0 \Leftrightarrow T$ not funcⁿ of S

• Let S be complete and suff. stat. & T be an unb. est. of θ .

Theorem: If T is an unb. est. of θ , and S is complete suff., then $T^* = E(T|S)$ is UMVUE.

• This is the Lehmann-Scheffe theorem.

Proof: # Exercise

Ex1: $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Bern}(\theta)$, $\theta \in (0,1)$

$\hat{\theta} = \bar{X}$ = sample proportion is UMVUE for θ .
Because $\hat{\theta}$ is suff. as well as complete, as well as
unb. on unb. est. of θ , by LS-theorem $\hat{\theta}$ is
THE UMVUE.

Ex2: $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Poi}(\theta)$ $\forall \theta > 0$

$$S = \sum_{i=1}^n X_i, \quad S \sim \text{Poi}(n\theta)$$

S is suff. as well as complete.
 $\therefore T = \frac{S}{n}$ is THE UMVUE by virtue of LS.

Ex3: $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Unif}[0, \theta]$, $\theta > 0$

~~It is suff. Is it complete?~~

Let $S = X_{(n)}$. S is suff. (we can check)

a) Is S complete?

A) Apparently it is, but we can't prove it yet. ☹

$\therefore T = \frac{n+1}{n} S$ is THE UMVUE.

Ex4) $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Expo}(\theta)$

$S = \sum_{i=1}^n X_i \sim \text{Gamma}(n, \theta)$ is suff.

~~for completeness~~ It is also complete, but, as
you can tell, we can't prove how. So, S is
complete (ON GOD) & $T = \frac{S}{n}$ is THE UMVUE.

Theorem: $X_1, \dots, X_n \stackrel{iid}{\sim} f_\theta(x)$ (pdf or pmf).

$$\text{Joint pdf / pmf} := \prod_{i=1}^n f_\theta(x_i) \stackrel{\text{exp.}}{=} h(x_1, \dots, x_n) \cdot g(\theta, s(x_1, \dots, x_n))$$

$h \rightarrow$ free of θ ; $g \rightarrow$ depends on $x_1, \dots, x_n$ through s .

Then, $s(X_1, \dots, X_n)$ is suff. stat.

• This is called Neymann-Fisher factorisation.

Ex3: $X_1, \dots, X_n \stackrel{iid}{\sim} U[0, \theta]$

$$f_\theta(x) = \frac{1}{\theta} I[x \in (0, \theta)]$$

$$\text{NF factorization: } \prod_{i=1}^n f_\theta(x_i) = \frac{1}{\theta^n} \prod_{i=1}^n I[0 < x_i < \theta]$$

$$= \frac{1}{\theta^n} \cdot I[X_{(1)} > 0] \cdot I[X_{(n)} < \theta]$$

$$= \left(\frac{I[X_{(n)} < \theta]}{\theta^n} \right) \cdot I[X_{(1)} > 0]$$

$\downarrow$ $\downarrow$
 g h

$\therefore I[X_{(n)} < \theta]$ is a suff. stat.

Identify suff. stat. in Ex 1, 2 & 4 using NF factorizⁿ.

Ex5: $X_1, \dots, X_n \stackrel{iid}{\sim} N(\theta, 1)$, $\theta \in \mathbb{R}$

$$f_\theta(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-\theta)^2}{2}}$$

$$\therefore \prod_{i=1}^n f_\theta(x_i) = \frac{1}{(2\pi)^{\frac{n}{2}}} e^{-\frac{1}{2} \sum_{i=1}^n (x_i - \theta)^2} = \frac{\exp(-\frac{1}{2} \sum_{i=1}^n x_i^2) \cdot \exp(\theta \sum_{i=1}^n x_i - \frac{n}{2} \theta^2)}{(2\pi)^{\frac{n}{2}}}$$

$$= \underbrace{\exp\left(-\frac{1}{2} \sum_{i=1}^n x_i^2\right)}_{h} \cdot \underbrace{\exp\left[\theta \sum_{i=1}^n x_i - \frac{n}{2} \theta^2\right]}_{g}$$

$\therefore S = \sum_{i=1}^n x_i$ is suff.

$$S \sim N(n\theta, n)$$

In a similar way, as prev., S is also complete.
 $\therefore T = S/n$ is THE UMVUE.

Ex6: $X_1, \dots, X_n \stackrel{iid}{\sim} \frac{1}{\beta(\theta, \phi)} x^{\theta-1} (1-x)^{\phi-1}, \theta, \phi > 0, 0 < x < 1.$

$$\begin{aligned} \prod_{i=1}^n f_{\theta, \phi}(x_i) &= \frac{1}{\beta^n(\theta, \phi)} (x_1 x_2 \dots x_n)^{\theta-1} \cdot [(1-x_1) \dots (1-x_n)]^{\phi-1} \\ &= \underbrace{\left[\left(\prod_{i=1}^n x_i \right)^{-1} \left(\prod_{i=1}^n (1-x_i) \right)^{-1} \right]}_h \left[\frac{(x_1 \dots x_n)^{\theta} [(1-x_1) \dots (1-x_n)]^{\phi}}{\beta^n(\theta, \phi)} \right] \end{aligned}$$

$\therefore S = \left(\prod_{i=1}^n x_i, \prod_{i=1}^n (1-x_i) \right)$ is a suff. stat.

Ex7: $X_1, \dots, X_n \stackrel{iid}{\sim} \frac{1}{\pi(1+(x-\theta)^2)}$

$$\begin{aligned} \prod_{i=1}^n f_{\theta}(x_i) &= \pi^{-n} \prod_{i=1}^n (1+(x_i-\theta)^2)^{-1} \\ &= \pi^{-n} \prod_{i=1}^n [(1+(x_i)-\theta)^2]^{-1} \end{aligned}$$

$\therefore S = X_{(n)}$ is a suff. stat.

* Order statistic is ALWAYS a suff. stat. for data having pdf.